Numerical Analysis I
Quiz 8A — Solutions

Question 1

Consider the function

$$f(x) = \frac{1}{x}, \qquad x > 0.$$

We want to approximate $f(x)$ using polynomial interpolation at the four nodes

$$x_0 = 1, \qquad x_1 = 2, \qquad x_2 = 3, \qquad x_3 = 4.$$

(a) Complete the following Newton divided difference table. The blank entries are to be computed.

Using

$$f[x_i, \ldots, x_{i+k}] = \frac{f[x_{i+1}, \ldots, x_{i+k}] - f[x_i, \ldots, x_{i+k-1}]}{x_{i+k} - x_i},$$

we obtain

x_i	$f[x_i]$	First differences	Second differences	Third differences
1	1	$-\frac{1}{2}$	$\frac{1}{6}$	$-\frac{1}{24}$
2	$\frac{1}{2}$	$-\frac{1}{6}$	$\frac{1}{24}$	
3	$\frac{1}{3}$	$-\frac{1}{12}$		
4	$\frac{1}{4}$			

In particular,

$$f[2,3,4] = \frac{-1/12 + 1/6}{4-2} = \frac{1}{24}, \qquad f[1,2,3,4] = \frac{1/24 - 1/6}{4-1} = -\frac{1}{24}.$$

(b) Using the divided differences from part (a), write the polynomial interpolant $p_3(x)$ of degree at most 3 in Newton's nested form.

Newton's nested form gives

$$\boxed{p_3(x) = 1 + (x-1)\left[-\frac{1}{2} + (x-2)\left(\frac{1}{6} - \frac{1}{24}(x-3)\right)\right].}$$

(c) Use the polynomial interpolation error theorem to find an upper bound for $|f(2.5) - p_3(2.5)|$.

The interpolation error theorem states that, for some $\xi \in (1,4)$,

$$f(2.5) - p_3(2.5) = \frac{f^{(4)}(\xi)}{4!}(2.5-1)(2.5-2)(2.5-3)(2.5-4).$$

Since

$$f^{(4)}(x) = \frac{24}{x^5}, \qquad \max_{x\in[1,4]} |f^{(4)}(x)| = 24,$$

we obtain

$$\boxed{|f(2.5) - p_3(2.5)| \le \frac{24}{24}\left|\frac{3}{2}\cdot\frac{1}{2}\cdot\left(-\frac{1}{2}\right)\cdot\left(-\frac{3}{2}\right)\right| = \frac{9}{16}.}$$

Question 2

(a) What is the definition of the Chebyshev polynomials of the first kind $T_n(x)$?

For $x \in [-1, 1]$, the Chebyshev polynomial of the first kind is defined by

$$\boxed{T_n(x) = \cos(n \arccos x), \qquad n = 0, 1, 2, \ldots.}$$

Equivalently, $T_n(\cos\theta) = \cos(n\theta)$. In particular, $T_0(x) = 1$ and $T_1(x) = x$.

(b) Derive the three-term recurrence relation for the Chebyshev polynomials T_n.

The trigonometric identity

$$\cos((n+1)\theta) + \cos((n-1)\theta) = 2\cos\theta\cos(n\theta)$$

with $x = \cos\theta$ implies

$$T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x).$$

Thus,

$$\boxed{T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \qquad n \ge 1,}$$

with $T_0(x) = 1$ and $T_1(x) = x$.

(c) We are given $T_3(x) = 4x^3 - 3x$ and $T_4(x) = 8x^4 - 8x^2 + 1$. Use the recurrence relation from part (b) to compute $T_5(x)$.

By the recurrence,

$$\begin{aligned} T_5(x) &= 2xT_4(x) - T_3(x) \\ &= 2x(8x^4 - 8x^2 + 1) - (4x^3 - 3x) \\ &= \boxed{16x^5 - 20x^3 + 5x}. \end{aligned}$$

(d) Find all the zeros of $T_5(x)$.

By the defining identity, $T_5(\cos\theta) = \cos(5\theta)$. The five zeros in $[-1, 1]$ therefore satisfy

$$5\theta = \frac{(2k-1)\pi}{2}, \qquad k = 1, \ldots, 5.$$

Hence all five zeros are

$$\boxed{x_k = \cos\left(\frac{(2k-1)\pi}{10}\right), \qquad k = 1, \ldots, 5.}$$

In increasing order, they are

$$-\cos\frac{\pi}{10}, \quad -\cos\frac{3\pi}{10}, \quad 0, \quad \cos\frac{3\pi}{10}, \quad \cos\frac{\pi}{10}.$$

(e) Sketch the graph of $T_5(x)$ on $[-1, 1]$. Clearly indicate its zeros and its values at the points where it attains 1 or -1.

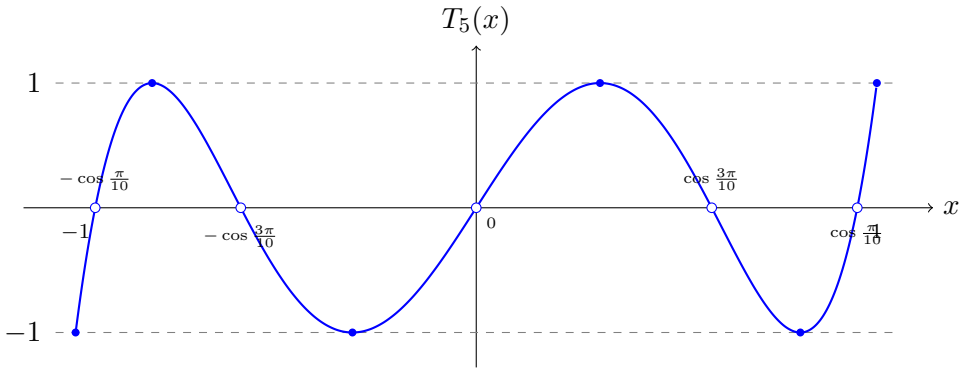
The extrema occur at

$$x = \cos\left(\frac{k\pi}{5}\right), \qquad k = 0, \ldots, 5,$$

where

$$T_5\left(\cos\frac{k\pi}{5}\right) = (-1)^k.$$

The following sketch labels the five zeros with open circles and the six extrema with filled dots.



In increasing order of x, the extrema are

x	-1	$-\cos(\pi/5)$	$-\cos(2\pi/5)$	$\cos(2\pi/5)$	$\cos(\pi/5)$	1
$T_5(x)$	-1	1	-1	1	-1	1

(f) State (without proof) the minimax property of the Chebyshev polynomial T_5. In other words, explain what makes T_5 special among polynomials of degree 5 with the same leading coefficient.

Since T_5 has leading coefficient 16, it minimizes the maximum absolute value on $[-1, 1]$ among all degree-5 polynomials with leading coefficient 16. More precisely,

$$\boxed{\max_{x\in[-1,1]} |p(x)| \geq \max_{x\in[-1,1]} |T_5(x)| = 1}$$

for every polynomial p of degree 5 with leading coefficient 16, with equality if and only if $p = T_5$.